

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

GCP vs AWS — Mapare Completa a Serviciilor

GCP Associate Cloud Engineer

Study Notes Study Notes April 2026

Toate serviciile GCP mentionate in materialele de studiu ACE, cu echivalentul lor din AWS.

Compute

GCP	AWS	Descriere
Compute Engine	EC2	Masini virtuale (IaaS)
Spot VMs	Spot Instances	VM-uri cu discount mare, pot fi revendicate oricand
Preemptible VMs	Spot Instances (legacy)	Predecesorul Spot VMs, max 24h
Sole-Tenant Nodes	Dedicated Hosts	Servere fizice dedicate
Custom Machine Types	— (nu exista direct)	CPU/RAM configurabil exact; AWS are doar tipuri predefinite
Instance Templates	Launch Templates	Blueprint imutabil pentru creare VM-uri
Managed Instance Groups (MIG)	Auto Scaling Groups (ASG)	Grup de VM-uri identice cu autoscaling si autohealing
OS Login	EC2 Instance Connect / SSM	Acces SSH bazat pe identitate (IAM)
VM Manager	AWS Systems Manager (SSM)	Patch management, inventar OS, configurare
Google Kubernetes Engine (GKE)	Elastic Kubernetes Service (EKS)	Kubernetes gestionat
GKE Autopilot	EKS with Fargate	Kubernetes serverless (platesti per pod)
GKE Enterprise (Anthos)	EKS Anywhere	Kubernetes hybrid/multi-cloud
Cloud Run	AWS App Runner / Fargate	Containere serverless (scale to zero)
Cloud Functions	AWS Lambda	Functions as a Service (FaaS)
Cloud Build	AWS CodeBuild	Build si CI/CD pentru containere

Storage

GCP	AWS	Descriere
Cloud Storage	S3	Object storage
Cloud Storage — Standard	S3 Standard	Date accesate frecvent
Cloud Storage — Nearline	S3 Standard-IA	Date accesate < 1x/luna
Cloud Storage — Coldline	S3 Glacier Instant Retrieval	Date accesate < 1x/trimestru
Cloud Storage — Archive	S3 Glacier Deep Archive	Date accesate < 1x/an
Persistent Disk (pd-standard)	EBS gp2/gp3 (HDD: st1/sc1)	Block storage — HDD
Persistent Disk (pd-balanced)	EBS gp3	Block storage — SSD echilibrat
Persistent Disk (pd-ssd)	EBS io1/io2	Block storage — SSD performant
Persistent Disk (pd-extreme)	EBS io2 Block Express	Block storage — IOPS maxim
Regional Persistent Disk	EBS Multi-Attach (limitat)	Block storage replicat in 2 zone
Local SSD	Instance Store (NVMe)	SSD efemer, atasat fizic
Filestore	EFS (Elastic File System)	NFS managed file storage
Storage Transfer Service	AWS DataSync	Transfer date la scara mare
Transfer Appliance	AWS Snowball / Snowball Edge	Dispozitiv fizic pentru transfer offline

Databases

GCP	AWS	Descriere
Cloud SQL (MySQL)	RDS MySQL / Aurora MySQL	MySQL gestionat
Cloud SQL (PostgreSQL)	RDS PostgreSQL / Aurora PostgreSQL	PostgreSQL gestionat
Cloud SQL (SQL Server)	RDS SQL Server	SQL Server gestionat
AlloyDB	Aurora PostgreSQL	PostgreSQL de inalta performanta (HTAP)
Cloud Spanner	Aurora Global Database (partial) / DynamoDB Global Tables	Baza de date relationala, globala, strong consistency
BigQuery	Amazon Redshift / Athena	Data warehouse serverless (OLAP)
BigQuery ML	Amazon SageMaker (partial)	ML direct in SQL
Firestore (Native mode)	DynamoDB + AppSync	Document DB cu real-time sync
Firestore (Datastore mode)	DynamoDB	Document DB pentru server-side
Cloud Bigtable	DynamoDB / Amazon Timestream	Wide-column NoSQL, time-series
Memorystore (Redis)	ElastiCache (Redis)	Redis gestionat in-memory
Memorystore (Memcached)	ElastiCache (Memcached)	Memcached gestionat

Networking

GCP	AWS	Descriere
VPC	VPC	Retea virtuala privata
Subnets	Subnets	Segmente IP regionale
Firewall Rules	Security Groups + NACLs	Reguli de trafic (GCP: la nivel de VPC, stateful)
Routes	Route Tables	Reguli de rutare trafic
Cloud NAT	NAT Gateway	Acces internet outbound pentru VM-uri fara IP extern
Cloud DNS	Route 53	DNS gestionat (public si privat)
Cloud CDN	CloudFront	Content Delivery Network
Cloud Armor	AWS WAF + Shield	Protectie DDoS si Web Application Firewall
External Application LB (HTTP/S)	Application Load Balancer (ALB)	Load balancer L7, global
Internal Application LB	Internal ALB	Load balancer L7, intern
External Passthrough Network LB	Network Load Balancer (NLB)	Load balancer L4, TCP/UDP
Internal Passthrough Network LB	Internal NLB	Load balancer L4, intern
External Proxy Network LB	NLB with TLS termination	Load balancer L4, TCP/SSL proxy
Cloud VPN (HA VPN)	AWS Site-to-Site VPN	VPN IPsec catre on-premises
Cloud Interconnect (Dedicated)	AWS Direct Connect	Conexiune fizica dedicata
Cloud Interconnect (Partner)	AWS Direct Connect via partner	Conexiune prin provider

GCP	AWS	Descriere
VPC Network Peering	VPC Peering	Conectare VPC-uri prin IP intern
Shared VPC	AWS RAM (Resource Access Manager)	VPC partajat intre proiecte/conturi
Private Google Access	VPC Endpoints (Gateway type)	Acces la API-uri Google fara IP extern
Serverless VPC Access Connector	— (Fargate uses VPC natively)	Conecteaza Cloud Run/Functions la VPC
Cloud Router	— (BGP inclus in VPN/DX)	BGP router pentru rutare dinamica
Network Service Tiers (Premium)	— (default AWS routing)	Trafic prin backbone Google
Network Service Tiers (Standard)	— (similar cu default AWS)	Trafic prin internet public
Identity-Aware Proxy (IAP)	AWS SSM Session Manager	Acces securizat la VM-uri fara IP extern

Security si Identity

GCP	AWS	Descriere
Cloud IAM	AWS IAM	Identity and Access Management
IAM Roles (Basic)	AWS Managed Policies (broad)	Roluri largi: Viewer, Editor, Owner
IAM Roles (Predefined)	AWS Managed Policies	Roluri specifice per serviciu
IAM Roles (Custom)	IAM Custom Policies	Roluri definite de utilizator
IAM Conditions	IAM Condition Keys	Acces conditionat (timp, resursa, etc.)
IAM Deny Policies	SCPs (Service Control Policies)	Blocare explicita de permisiuni
Service Accounts	IAM Roles for Services / Instance Profiles	Identitate pentru aplicatii/VM-uri
Workload Identity (GKE)	IRSA (IAM Roles for Service Accounts)	Pod-uri K8s → identitate cloud
Workload Identity Federation	IAM OIDC Identity Providers	Identitati externe (GitHub, AWS) → GCP
Service Account Keys	Access Keys / Secret Keys	Credentiale long-lived (de evitat!)
Service Account Impersonation	AssumeRole (STS)	Impersonare temporara
Cloud Identity	AWS IAM Identity Center (SSO)	Managementul utilizatorilor si grupurilor
Organization	AWS Organizations	Nivel ierarhic de top
Folders	Organizational Units (OUs)	Grupare logica sub organizatie
Organization Policies	Service Control Policies (SCPs)	Constrangeri top-down pe resurse

GCP	AWS	Descriere
Secret Manager	AWS Secrets Manager	Stocare secrete (parole, chei API)
Cloud KMS	AWS KMS	Managementul cheilor de criptare
OS Login	EC2 Instance Connect	SSH bazat pe identitate IAM

Monitoring, Logging si Operations

GCP	AWS	Descriere
Cloud Monitoring	CloudWatch Metrics	Metriци, dashboarduri, alerte
Cloud Logging	CloudWatch Logs	Centralizare si analiza loguri
Cloud Trace	AWS X-Ray	Distributed tracing
Cloud Profiler	CodeGuru Profiler	Profilare CPU/memorie
Error Reporting	— (CloudWatch + X-Ray)	Agregare si afisare erori
Ops Agent	CloudWatch Agent	Agent unificat pentru metriци si loguri pe VM-uri
Managed Service for Prometheus	Amazon Managed Prometheus (AMP)	Prometheus gestionat
Log Router / Log Sinks	CloudWatch Subscription Filters	Rutare loguri catre destinatii
Log-Based Metrics	CloudWatch Metric Filters	Metriци derivate din loguri
Uptime Checks	Route 53 Health Checks / CloudWatch Synthetics	Verificare disponibilitate serviciu
Audit Logs (Admin Activity)	CloudTrail (Management Events)	Loguri API-uri de modificare — mereu activ, gratuit
Audit Logs (Data Access)	CloudTrail (Data Events)	Loguri API-uri de citire — optional, cu cost
Cloud Scheduler	EventBridge Scheduler	Cron jobs gestionat
Google Cloud Status Dashboard	AWS Health Dashboard	Starea serviciilor cloud

Data & Analytics

GCP	AWS	Descriere
Pub/Sub	Amazon SNS + SQS / Amazon Kinesis	Messaging async (topic/subscription)
Dataflow	Amazon Kinesis Data Analytics / EMR	Stream si batch processing (Apache Beam)
Dataproc	Amazon EMR	Hadoop/Spark gestionat
Eventarc	Amazon EventBridge	Rutare evenimente (100+ surse)

Infrastructure as Code si DevOps

GCP	AWS	Descriere
Terraform	Terraform (identic)	IaC multi-cloud — HashiCorp
Cloud Foundation Toolkit (CFT)	AWS Landing Zone / Control Tower	Module Terraform cu best practices
Config Connector	AWS Controllers for K8s (ACK)	Resurse cloud ca obiecte K8s
Helm	Helm (identic)	Package manager pentru Kubernetes
Artifact Registry	Amazon ECR (Elastic Container Registry)	Registry pentru containere si pachete
Cloud Source Repositories	AWS CodeCommit	Git repository gestionat

Billing si Cost Management

GCP	AWS	Descriere
Billing Account	AWS Account (Payer Account)	Contul care plateste
Billing Budgets & Alerts	AWS Budgets	Bugete cu notificari
Billing Export (BigQuery)	Cost and Usage Reports (CUR) → Athena	Export detaliat costuri
Committed Use Discounts (CUDs)	Reserved Instances / Savings Plans	Discounturi pe 1-3 ani
Sustained Use Discounts (SUDs)	— (nu exista echivalent exact)	Discount automat > 25% din luna
Pricing Calculator	AWS Pricing Calculator	Estimare costuri
Labels	Tags	Perechi cheie-valoare pentru organizare si cost allocation
Quotas	Service Quotas (Limits)	Limite pe resurse per proiect/cont

Proiecte si Organizare

GCP	AWS	Descriere
Organization	AWS Organization	Nod radacina
Folder	Organizational Unit (OU)	Grupare ierarhica
Project	AWS Account	Unitatea de baza (resurse, billing, IAM)
Resource Hierarchy	AWS Organizations Hierarchy	Org → OU → Account

Diferente Cheie de Retinut

Aspect	GCP	AWS
VPC	Global (acoperă toate regiunile)	Regional (per regiune)
Subnets	Regionale (acoperă toate zonele din regiune)	Per zona de disponibilitate
Firewall	La nivel de VPC, doar allow/deny	Security Groups (allow only) + NACLs (allow/deny)
Load Balancing	Global nativ (L7)	Regional by default (Global cu CloudFront)
IAM	Roluri legate de resurse (binding)	Politici atasate utilizatorilor/grupurilor
Projects vs Accounts	Un proiect = izolare; mai multe proiecte sub un billing account	Un cont = izolare; Organizations grupeaza conturi
Spot VMs	Pot fi revendicate oricand, fara limita de timp	Spot Instances — similar, cu 2 min avertizare
Sustained Use Discounts	Automate, fara commitment	Nu exista — trebuie Reserved Instances explicit
Storage classes	Toate au aceeasi latenta (ms)	Glacier are latenta de ore
Serverless containers	Cloud Run (scale to zero, pay per request)	App Runner (similar) / Fargate (fara scale to zero)
K8s managed	GKE (cel mai matur K8s managed)	EKS (necesita mai multa configurare)
Database globala	Spanner (SQL + global strong consistency)	Nu exista echivalent direct

